

ABHI THIAGARAJAN

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EDUCATION

The University of Texas at Austin Bachelor of Science, Biomedical Engineering May 2028
Minor: Programming and Computation • Overall GPA: 4.00 /4.00

Relevant Coursework DSA, Introduction to Computing (ECE + BME), Linear Algebra, Differential Equations

EXPERIENCE

University of Texas at Dallas – Research Intern; Dallas, TX Aug 2025 – May 2026

- Designed and shipped a modular FastAPI service handling multi-modal medical imaging requests — built validation middleware, async request queuing, and a pluggable preprocessing layer supporting 3 scan types.
- Profiled and resolved concurrency bottlenecks in a distributed processing pipeline — restructured worker scheduling and batching logic, increasing throughput by 40% across concurrent queues.
- Built CLI automation tools that replaced manual pipeline execution steps, reducing operator intervention and improving end-to-end pipeline reliability across distributed environments.

360DigiTMG – Software Development Intern; Dallas, TX May 2025 – Aug 2025

- Engineered Python/SQL ETL pipelines ingesting 50K+ daily records, cutting data latency to near real-time.
- Built and deployed a production Flask REST API with versioned prediction endpoints, request validation, and automated regression benchmarking on holdout data — served live traffic.
- Built a self-serve internal dashboard platform eliminating 15+ hrs/week of ad-hoc analyst requests - reduced eng interrupt load.

Wolfram – Software Research Intern; Boston, MA May 2024 – August 2024

- Engineered distributed backend across 250K+ nodes- refactored sequential jobs into parallelized pipelines, cutting runtime by 60%.
- Built Python pipelines and real-time viz tooling for spatiotemporal simulation monitoring across distributed environments.
- Profiled and resolved memory and thread contention bottlenecks in GPU-accelerated molecular simulation pipelines — reduced peak memory usage by 35% on high-node workloads through targeted allocation and scheduling changes.

PROJECTS

CardioPulse – AI Cardiovascular Diagnostics Platform · Deployed March 2026

- Built and deployed a full-stack cardiovascular diagnostics platform — FastAPI backend serving real-time ECG analysis and cardiac risk prediction models, with a Plotly/Dash frontend validated with 5 clinical users.
- Deployed serverless inference on AWS Lambda + API Gateway — sub-second ECG response times for real-time clinical feedback.
- Designed clinician dashboards surfacing ECG metrics, risk scores, and predictions in a single view, reducing diagnostic lookup steps.

CarbonAI – Sustainability Optimization Platform · Deployed February 2026

- Built and deployed a full-stack sustainability platform — serverless Python backend and Dash frontend delivering real-time AI-generated carbon reduction recommendations based on user-submitted emissions data.
- Architected microservice APIs for LLM inference, carbon scoring, and personalized recommendation generation — designed for concurrent user sessions with stateless, horizontally scalable endpoints.

SubLeaseU - Sublease Marketplace for UT Students May 2026

- Engineered a full-stack sublease marketplace for UT students — designed normalized PostgreSQL schema, JWT auth with role-based API middleware, and RESTful Node.js/Express backend decoupled from a React frontend for independent deploys.

ACCOMPLISHMENTS

- WWT Hackathon (1st Place, ~200 teams) – built CNN-based diagnostic imaging classifier 2026
- EnviHacks - Sustainability Innovation Award + Girls Recode Hacks (2nd Place) 2026
- Wolfram Emerging Leaders/Scholarship Award – Top 10% of research papers 2025
- General Electric STEAM Lead Award - instructed 100+ students in Python, engineering, robotics 2025

SKILLS

Technical Skills: Java, C++, Python, JavaScript, SQL, REST APIs, Git, AWS (Lambda, API Gateway, S3), HTML/CSS

Certifications/Data Tools: AWS CCP, Azure Fundamentals, TensorFlow, Scikit-learn, Power BI

Organizations: Kappa Theta Pi (Pre-Professional Fraternity), Women in CS, Women in Engineering